

Bike Station Sharing System Analysis using Power BI

Project Overview and Objective

This project focuses on analyzing the **Bike Station Sharing System** dataset using **Power BI**. The raw data was cleaned, transformed, and modeled to create an interactive dashboard that provides meaningful insights into bike station availability, station capacity, contracts, and trends over time.

Objective

- Analyze bike station performance across different cities (contracts).
 - Monitor bike availability and station capacity.
 - Identify stations with low bike availability.
 - Track bike availability trends over time.
 - Support operational decision-making through interactive dashboards.
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2. Data Sources

- **Source Description:** Bike Sharing System Dataset (Google Dataset Search / Open Data)
 - **Timeline:** 2022–2025
 - **Domain:** Transportation / Smart City / Public Bike Sharing Analytics
 - **Dataset Type:** Bike Station Status Data
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3. Attribute (Column / Feature) Details

Attribute Name	Data Type	Description
Station Name	Text	Name of the bike station
Contract Name	Text	City/Contract operating the station
Available Bikes	Integer	Bikes currently available
Bike Stands	Integer	Total bike parking stands
Available Bike Stands	Integer	Empty bike stands available
Latitude	Decimal	Station latitude
Longitude	Decimal	Station longitude
Status	Text	OPEN/CLOSED station
Bonus	Boolean	Bonus station indicator
Date	Date	Station status date
Year	Integer	Year

Attribute Name	Data Type	Description
Month	Text	Month Name

4. Tools & Technologies

- **Power BI Desktop**
 - Power Query
 - Data Modeling
 - DAX Measures
 - Interactive Dashboard
 - Maps
 - Cards
 - Charts
 - Slicers
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5. Data Pre-Processing (Power Query)

Tasks Performed

- Removed duplicate records.
 - Checked and handled missing values.
 - Corrected data types.
 - Renamed columns.
 - Created Date table.
 - Added Year, Month, Quarter and Week Number.
 - Split dataset into Fact and Dimension tables.
 - Filtered unnecessary records.
 - Standardized text values.
 - Built Star Schema data model.
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6. Data Modelling and DAX (Power BI)

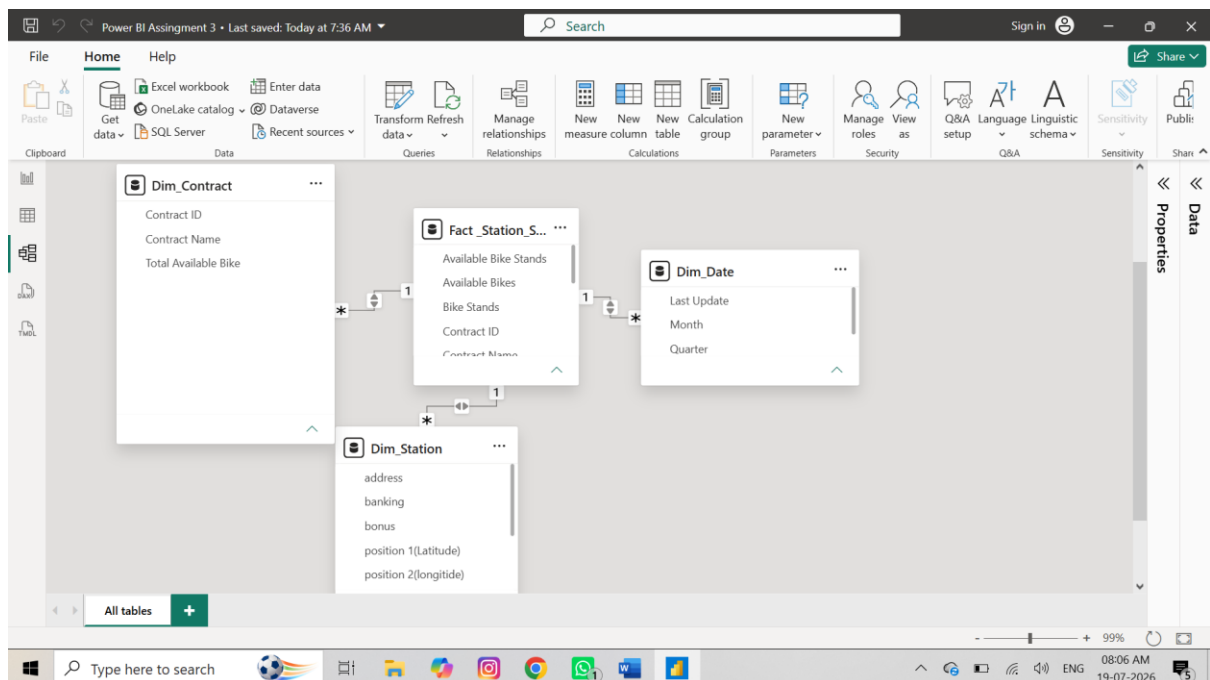
Data Model

Created relationships between:

- Dim_Station → Fact_Station_Status
- Dim_Contract → Fact_Station_Status
- Dim_Date → Fact_Station_Status

Relationship Type

- One-to-Many (1:*)



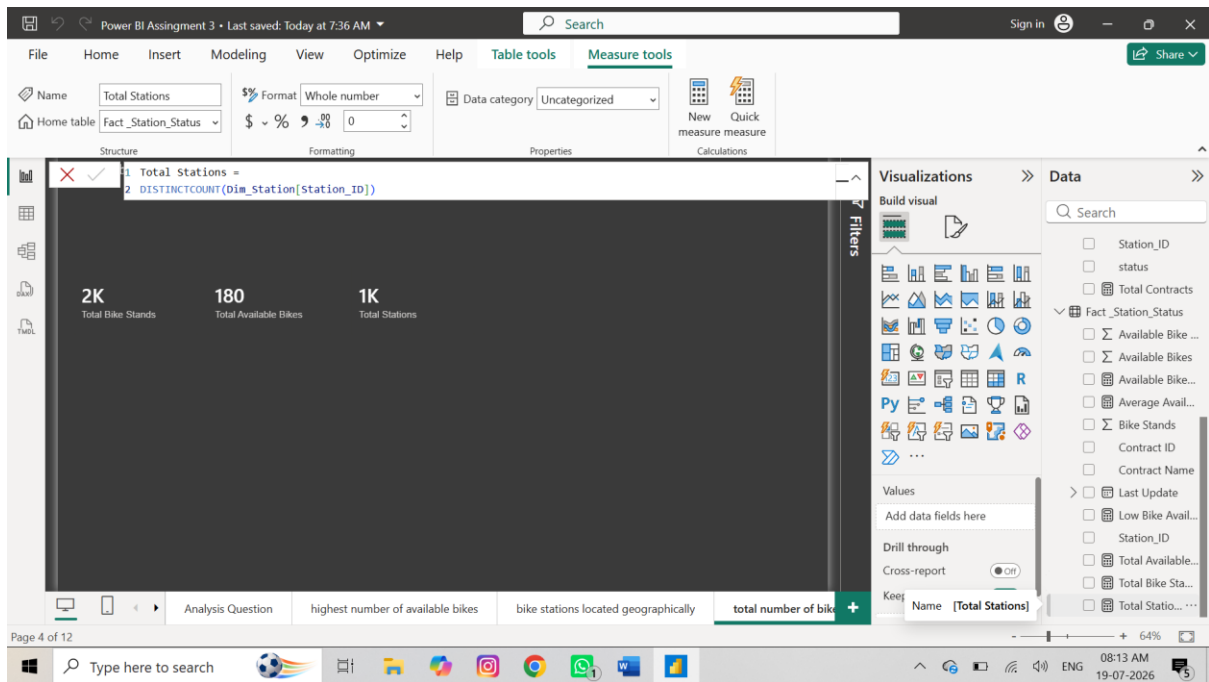
Calculated Columns (DAX)

Column Name	Formula
Year	<code>Year = YEAR(Fact_Station_Status[Date])</code>
Month	<code>Month = FORMAT(Fact_Station_Status[Date], "MMMM")</code>
Month Number	<code>Month No = MONTH(Fact_Station_Status[Date])</code>
Quarter	<code>Quarter = "Q" & FORMAT(Fact_Station_Status[Date], "Q")</code>

DAX Measures

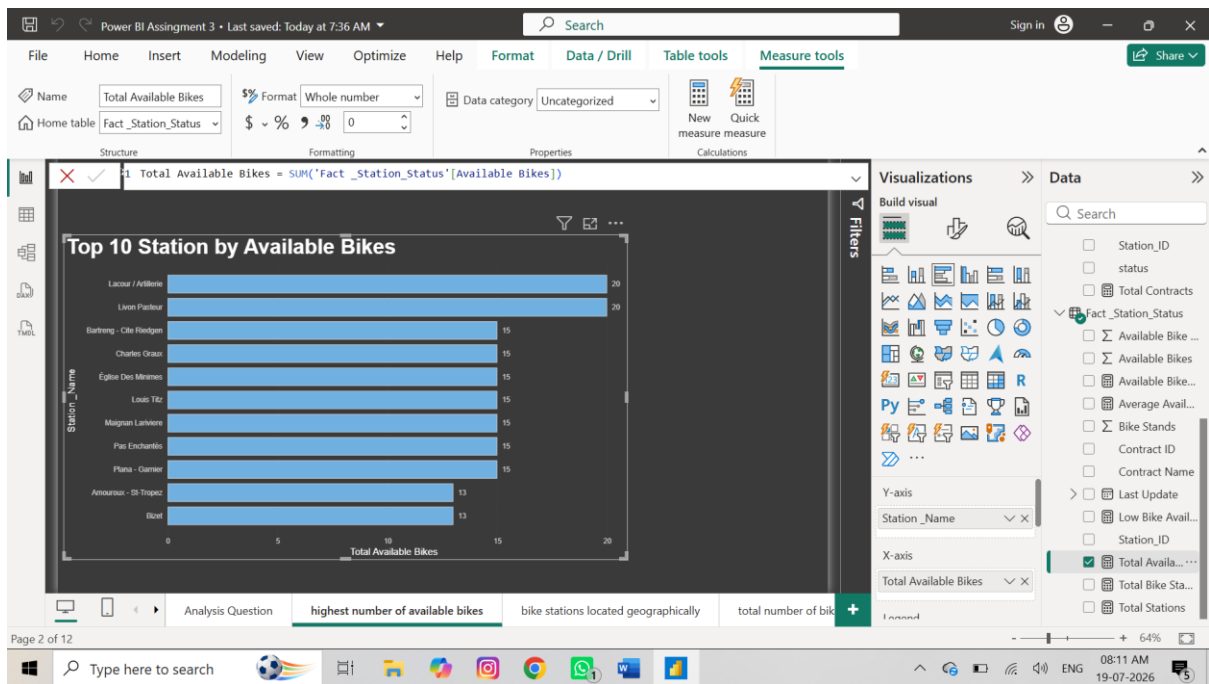
1. Total Stations

Total Stations =
`DISTINCTCOUNT(Dim_Station[Station_ID])`



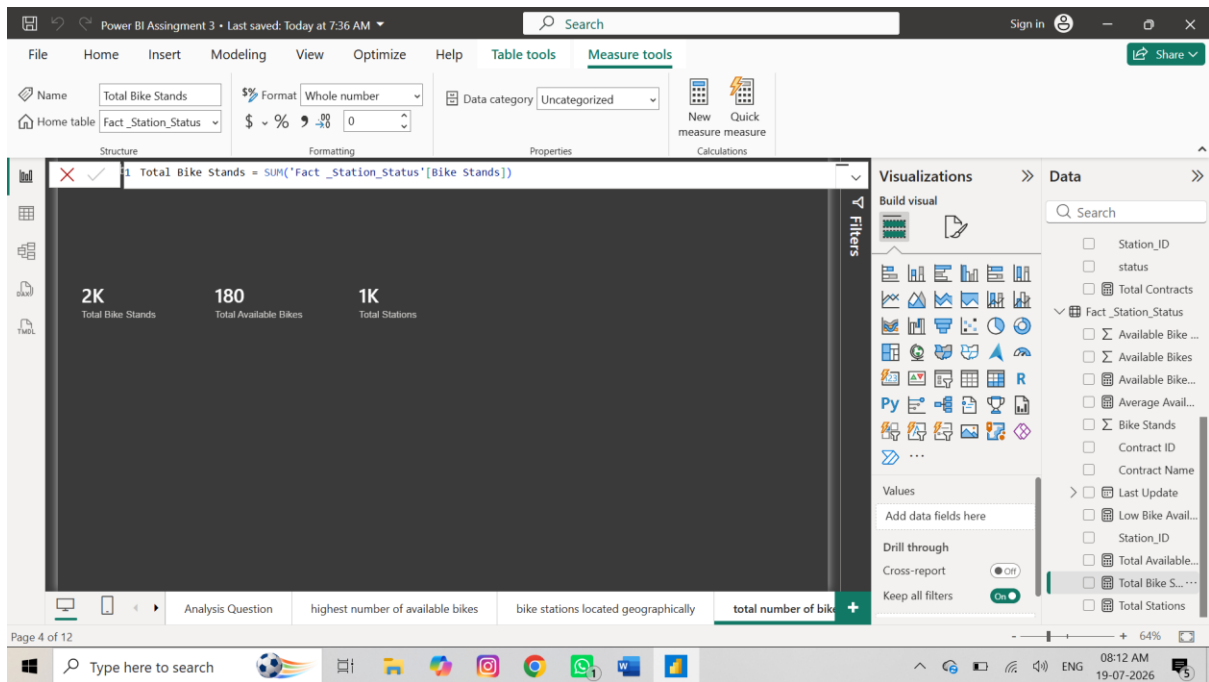
2. Total Available Bikes

Total Available Bikes =
 $\text{SUM}(\text{Fact_Station_Status}[\text{Available Bikes}])$



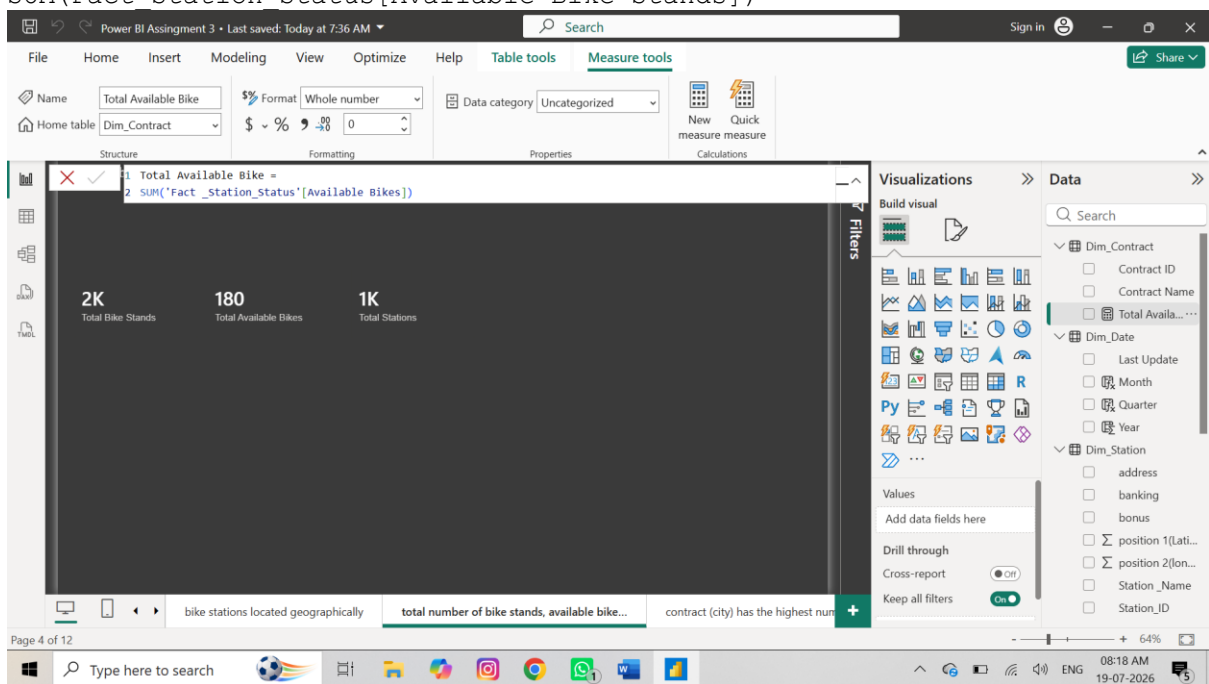
3. Total Bike Stands

Total Bike Stands =
 $\text{SUM}(\text{Fact_Station_Status}[\text{Bike Stands}])$



4. Total Available Bike Stands

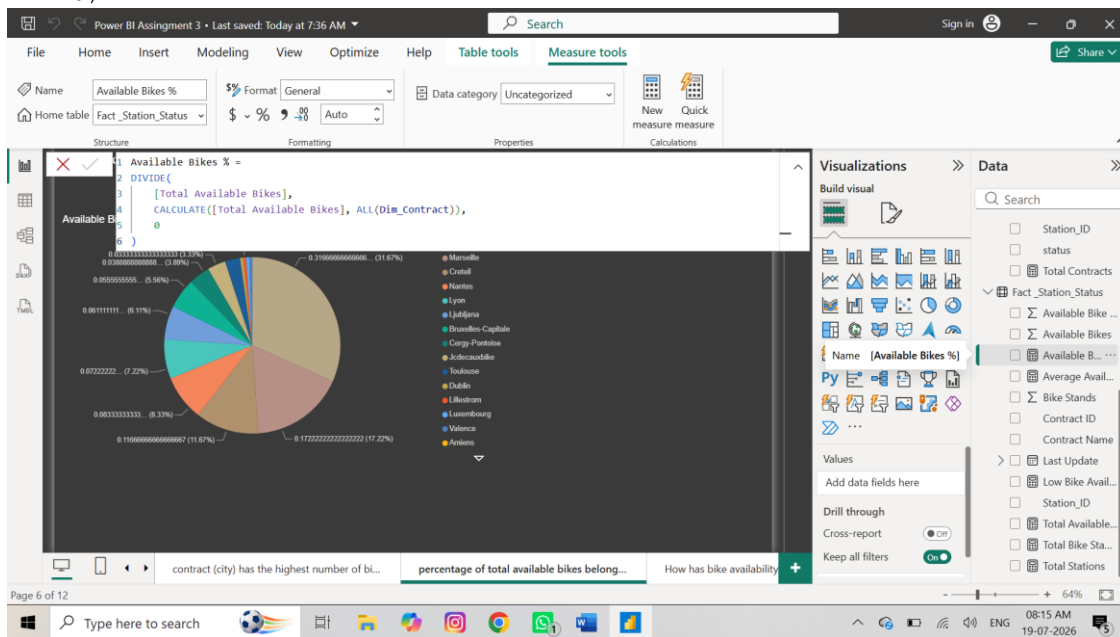
Total Available Bike Stands =
`SUM(Fact Station Status[Available Bike Stands])`



5. Available Bikes %

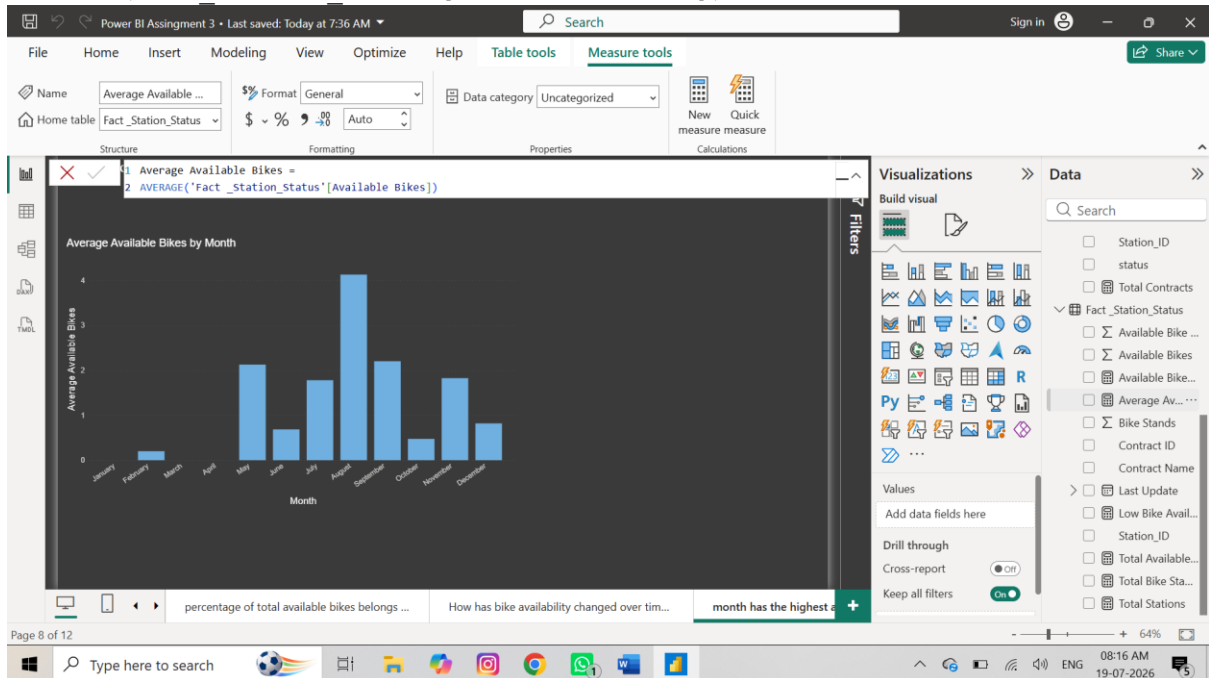
Available Bikes % =
`DIVIDE(`
`[Total Available Bikes],`

```
[Total Bike Stands],
0)
```



6. Average Available Bikes

```
Average Available Bikes =
AVERAGE(Fact_Station_Status[Available Bikes])
```



7. Low Bike Availability (Calculated Column)

```
Low Bike Availability =
IF(
```

)

7. Analysis and Visualizations (Power BI)

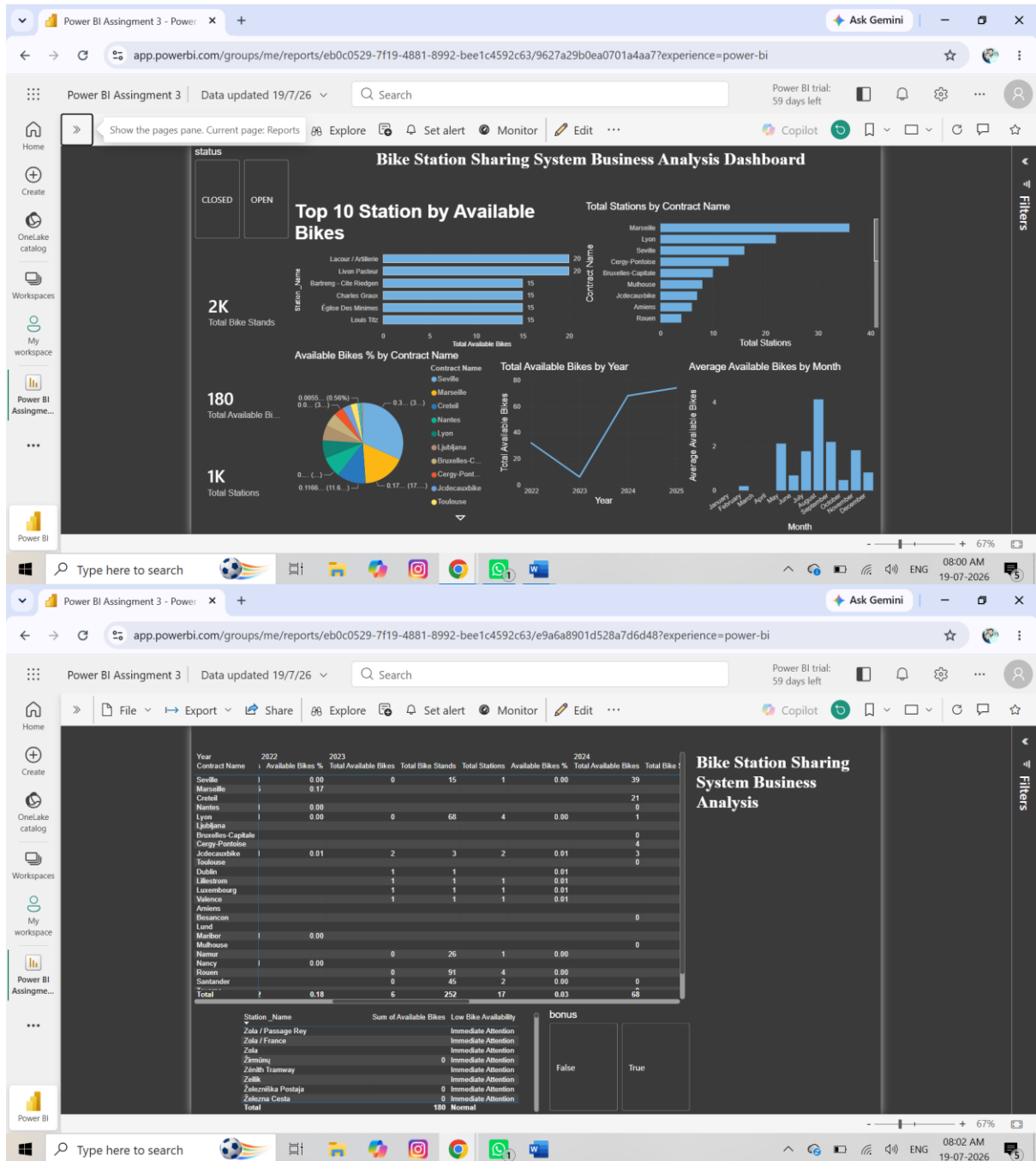
Dashboard Features

The dashboard contains the following visualizations:

1. Top 10 Stations by Available Bikes (Bar Chart)
2. Bike Station Geographic Locations (Map)
3. KPI Cards
 - o Total Bike Stands
 - o Total Available Bikes
 - o Total Stations
4. Total Stations by Contract (Bar Chart)
5. Available Bikes % by Contract (Pie/Donut Chart)
6. Bike Availability Trend by Year (Line Chart)
7. Average Available Bikes by Month (Column Chart)
8. Contract vs Bike Availability Matrix
9. Stations Requiring Immediate Attention (Table)
10. Slicers
 - o Year
 - o Contract
 - o Status
 - o Bonus

The dashboard also includes:

- Drill-down
- Filters
- Slicers
- Clear Titles
- Interactive Navigation



The dashboard answers questions such as top stations by available bikes, geographic distribution, contract comparisons, yearly trends, monthly averages, and stations needing attention.

8. Insights & Conclusions

Key Findings

- Marseille has the highest number of bike stations among all contracts.
 - Stations such as Lacour / Artillerie and Livon Pasteur recorded the highest number of available bikes.
 - Total Bike Stands are approximately 2,000, with 180 available bikes and around 1,000 stations in the dataset.
 - Several stations were identified as requiring immediate attention because of very low bike availability.
 - Bike availability varies significantly across contracts and years.
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- Top-performing stations have the highest bike availability.
 - Several stations have zero or very low bike availability.
 - Bike stations are distributed across multiple cities.
 - Marseille has one of the largest numbers of bike stations.
 - Total available bikes and bike stands are displayed using KPI cards.
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- Low bike availability is concentrated in specific stations.
 - Some contracts consistently perform better than others.
 - Closed stations contribute to lower bike availability.
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- Stations with consistently low bike availability may experience shortages during peak demand.
 - High-demand contracts are likely to require additional bike redistribution.
 - Future demand can be estimated using yearly availability trends.
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- Rebalance bikes to low-availability stations.
 - Increase bike stands in high-demand areas.
 - Monitor closed stations and restore operations quickly.
 - Improve maintenance scheduling.
 - Optimize bike distribution based on historical trends.

Conclusion

The Bike Station Sharing System dashboard successfully transforms raw operational data into meaningful business insights. Power BI's data modeling, DAX calculations, and interactive visualizations enable effective monitoring of bike availability, station capacity, and contract performance. The dashboard supports better decision-making for bike redistribution, resource planning, and service optimization, making it a valuable tool for transportation and smart city management.

